



Music Slot Booking — Master System Manual

Complete Technical Architecture, Feature Specifications & End-to-End Operation Guide

 **TARGET AUDIENCE:** Academy Owner / System Administrator / Developer

PLATFORM
PWA + Node.js REST

DATABASE
SQLite3 (WAL Mode)

SECURITY
Salted SHA-256 Passcode

STUDIO HELPLINE
9848173025 / 8919545774

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1. System Architecture & Zero-Dependency Backend

Music Slot Booking is an ultra-fast Progressive Web App engineered with zero heavy framework bloat. It features pure Vanilla JavaScript on the frontend and a native Node.js HTTP server (`server.js`) with SQLite3 in WAL mode.

2. Dual-Role Authentication & DevTools Security

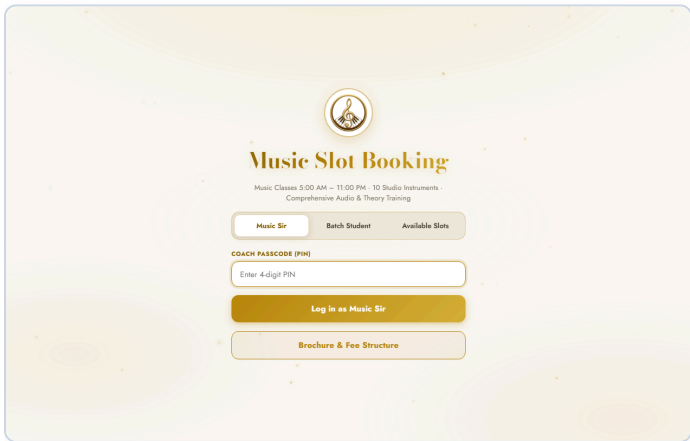


Fig 1: Dual Login Portal (Coach / Student / Free Slots)

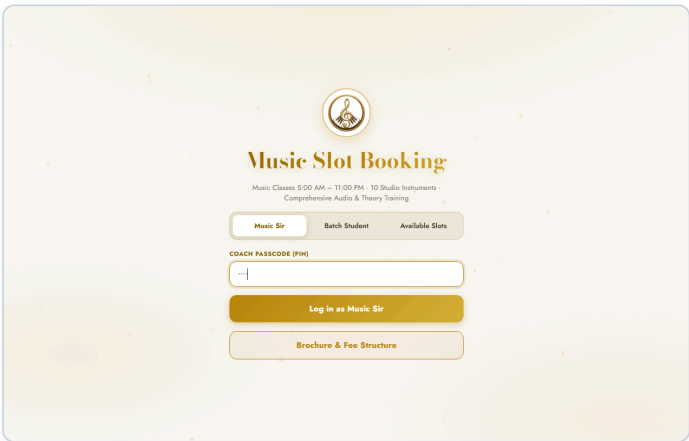


Fig 2: Coach Keypad with Salted SHA-256 Verification

Absolute DevTools Passcode Security

The Coach Passcode is **never hardcoded in client JavaScript, HTML, LocalStorage, or cookies**. When entered, it is sent over HTTPS to the server, where it is verified against a cryptographically salted SHA-256 hash. Even if someone inspects browser Developer Tools, they cannot access or extract Sir's passcode.

3. Music Sir (Coach) Portal & Daily Schedule (5 AM – 11 PM)

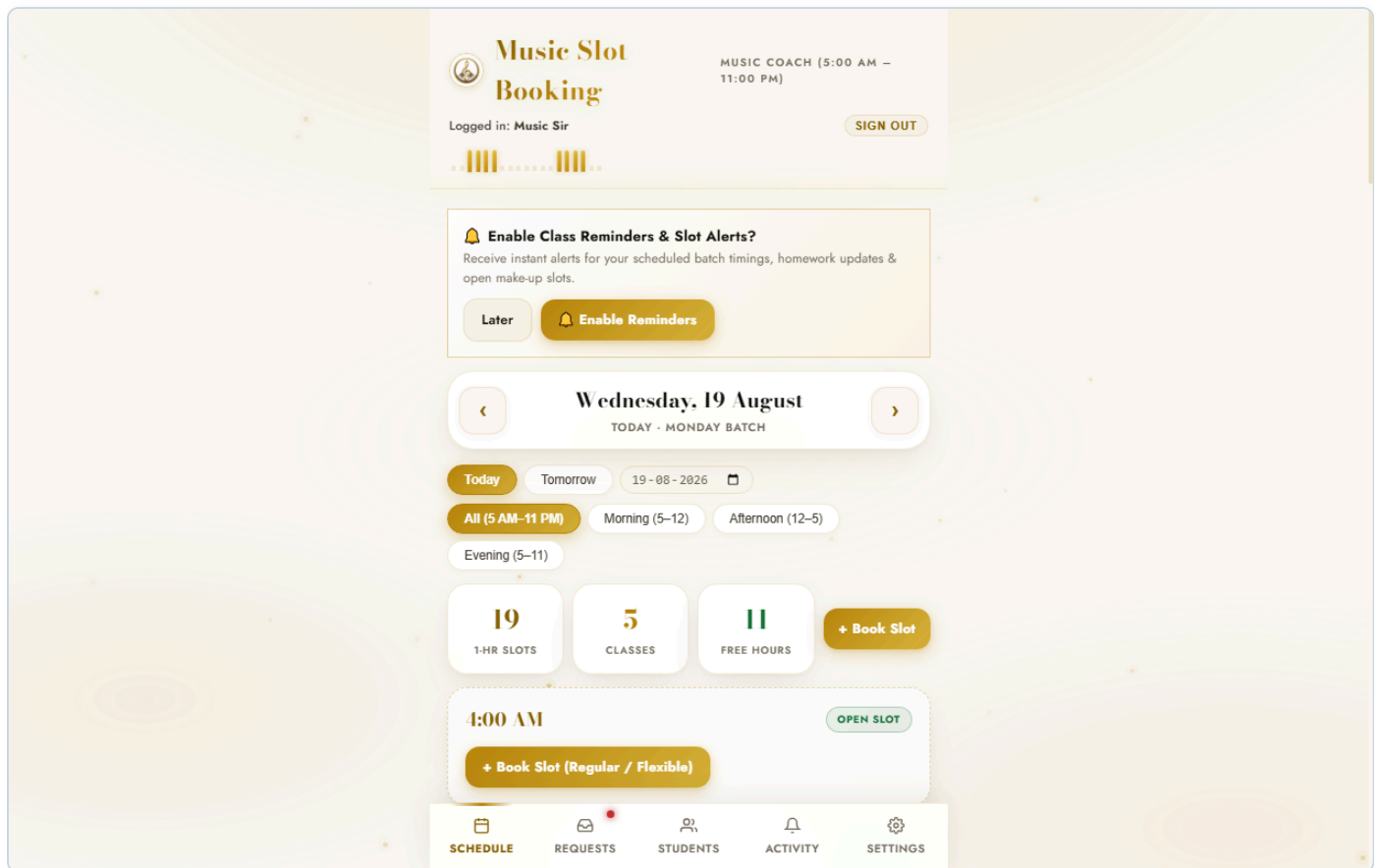


Fig 3: Daily Schedule Grid with Continuous Multi-Hour Spanning & Attendance Badges

The daily schedule displays all 19 hourly slots from **5:00 AM to 11:00 PM** for any selected date.

4. Attendance Engine & Auto Make-up Slot Recovery

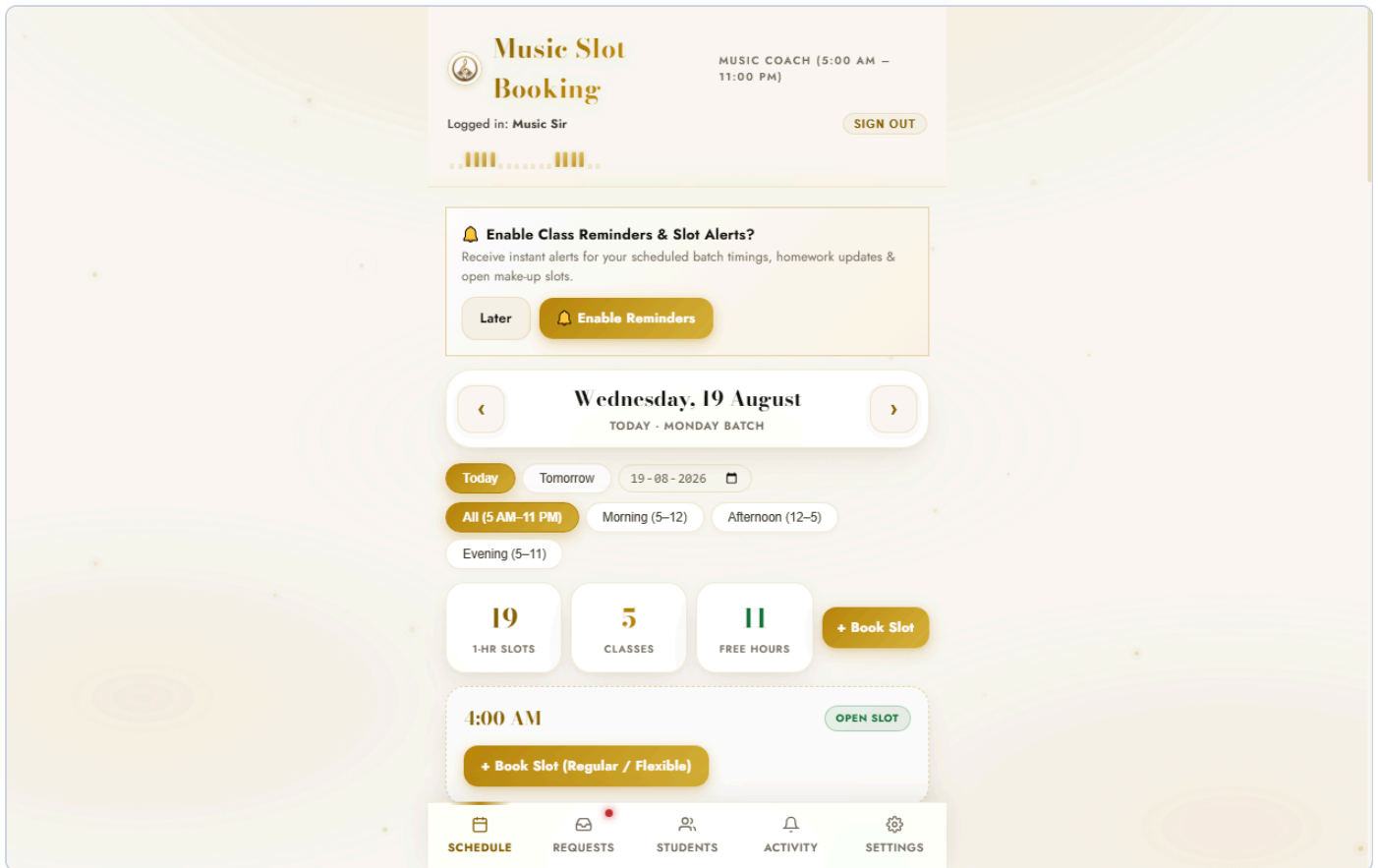


Fig 4: Sir [+ Book Slot] Manual Enrollment Sheet

When Sir marks any student [**X Absent**], the vacated slot is automatically transformed in real-time into an **Open Make-up Slot** for flexible students to claim!

5. Student Roster & Lesson Homework Management

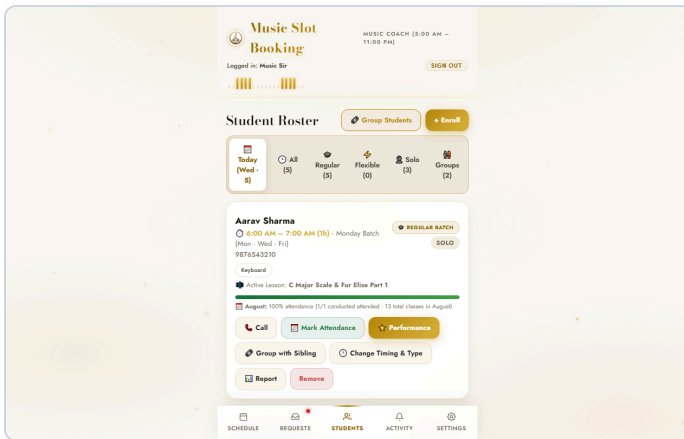


Fig 5: Student Roster with Instrument Badges

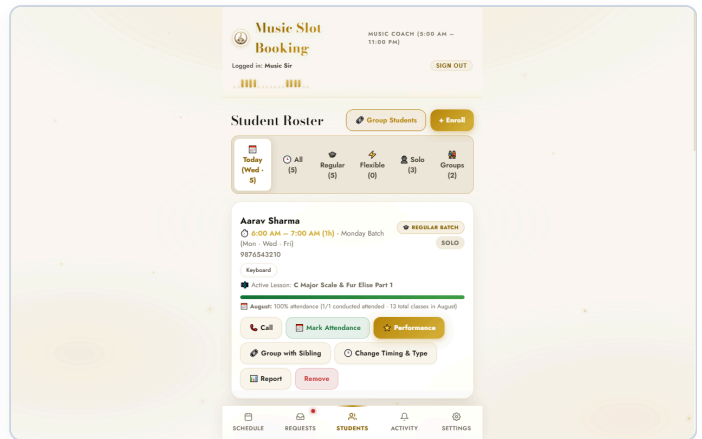


Fig 6: Lesson Progress & Homework Sheet

6. Student Join Request Approval & Batch Assignment

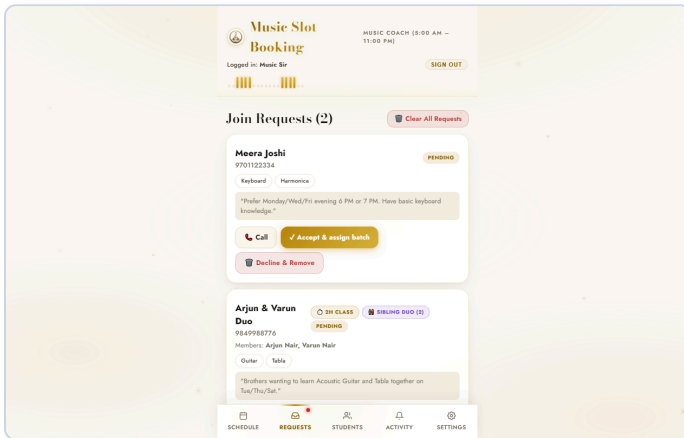


Fig 7: Pending Student Join Requests

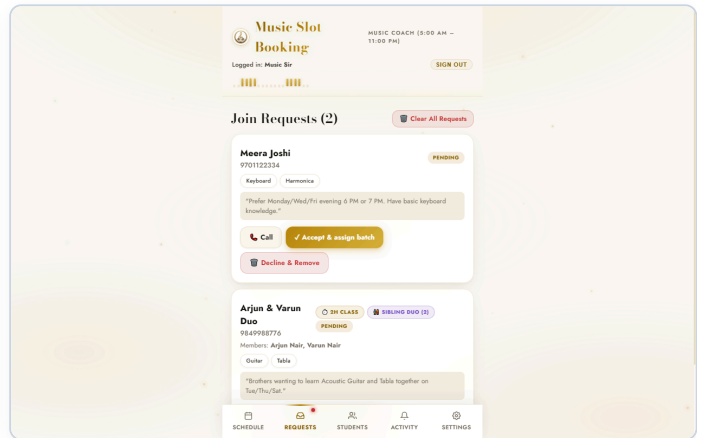


Fig 8: Smart Batch Slot Assigner Modal

7. Studio Settings, Hourly Slots (5 AM–11 PM) & Notifications

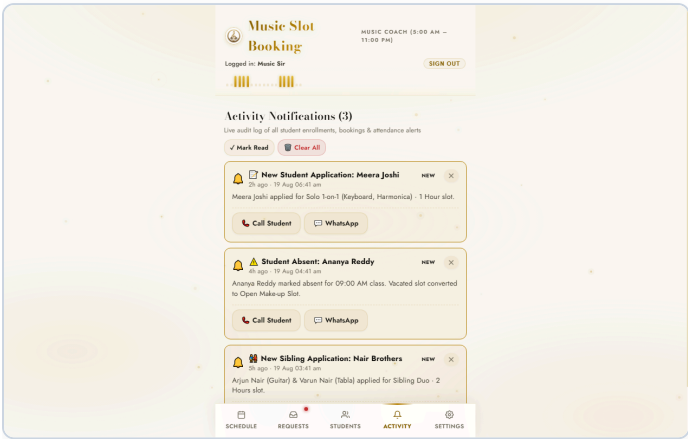


Fig 9: Activity Notifications with Clear All

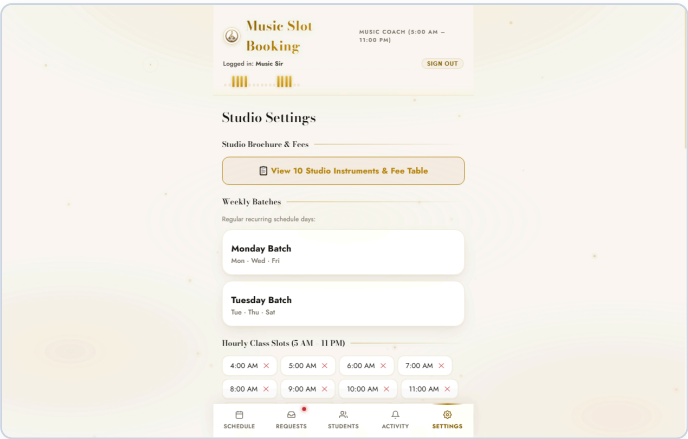


Fig 10: Studio Settings & Interactive Slots

8. Student Portal — Profile, Schedule & Homework

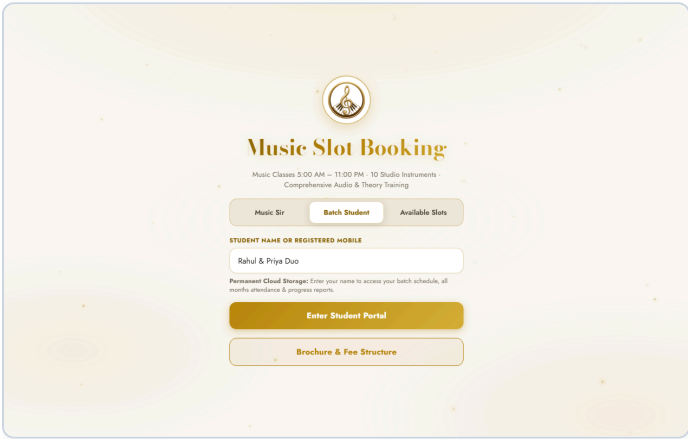


Fig 11: Student Search Login

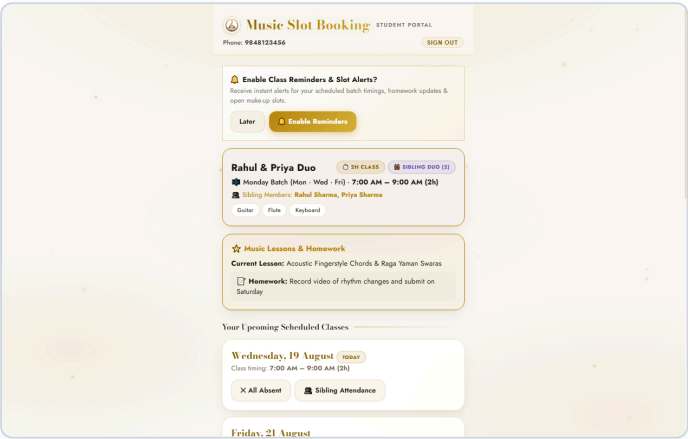


Fig 12: Enrolled Student Dashboard

9. Make-up Recovery & Flexible Booking Portals

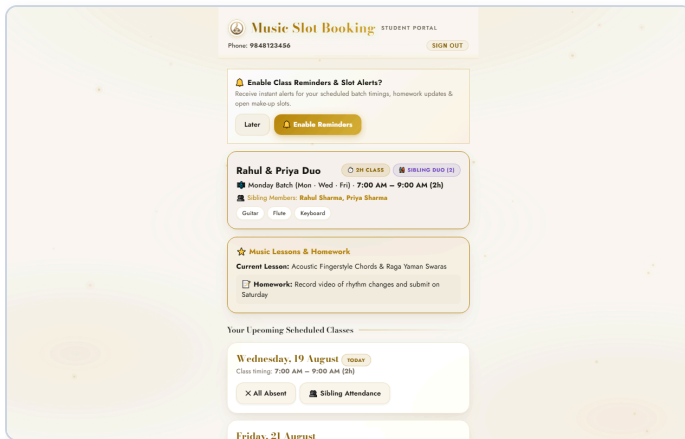


Fig 13: Claim Make-up Slot Modal

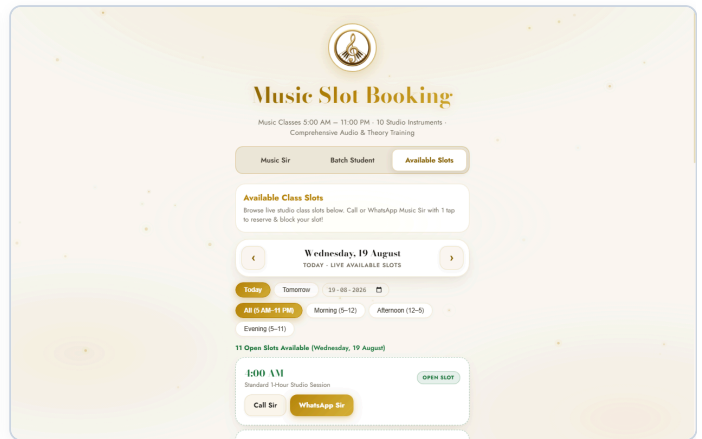


Fig 14: Available Slots Flexible Portal

10. How New Students Enroll (Solo vs. Sibling Step-by-Step)

The screenshot shows the 'Music Slot Booking' Student Portal. At the top, there's a header with the logo, name, and a 'SIGN OUT' button. Below the header, there's a section titled 'Join Music Classes' with a '10 Studio Instruments - Solo & Sibling Group Slots' text. Below that, there's a section titled 'View Official Fee Chart & Course Details' with a 'View Official Fee Chart & Course Details' button. Below that, there's a section titled 'Applicant Contact Information' with a 'YOUR FULL NAME' field containing 'Rohan Saxena' and a 'PRIMARY 10-DIGIT MOBILE NUMBER' field containing '10-digit mobile number'. Below that, there's a section titled 'CLASS SLOT FORMAT *' with a dropdown menu showing '- Select Class Slot Format -'. Below that, there's a section titled 'CLASS DURATION (HOURS PER CLASS) *' with a dropdown menu showing '- Select Duration (Hours per class) -'.

Fig 15: Initial Step-Locked State (Format & Duration start on -- Select --)

 Music Slot Booking

STUDENT PORTAL

Phone:

SIGN OUT



Join Music Classes

10 Studio Instruments - Solo & Sibling Group Slots

View Official Fee Chart & Course Details

 Applicant Contact Information

YOUR FULL NAME

PRIMARY 10-DIGIT MOBILE NUMBER

CLASS SLOT FORMAT *

 Individual Student (Solo 1-on-1)

CLASS DURATION (HOURS PER CLASS) *

 1 Hour (Standard Slot)

WHAT INSTRUMENT(S) DO YOU WANT TO LEARN? (SELECT FROM 10 INSTRUMENTS)

Keyboard

Guitar

Flute

Viola

Fig 16: Solo (10 Instruments Catalog Revealed)

 Music Slot Booking

STUDENT PORTAL

Phone:

SIGN OUT



Join Music Classes

10 Studio Instruments - Solo & Sibling Group Slots

View Official Fee Chart & Course Details

 Applicant Contact Information

YOUR FULL NAME

PRIMARY 10-DIGIT MOBILE NUMBER

CLASS SLOT FORMAT *

 Sibling Duo (2 Members / Brothers & Sisters)

CLASS DURATION (HOURS PER CLASS) *

 2 Hours (Double Slot — Recommended for Siblings)

FAMILY / SIBLING GROUP NAME (OPTIONAL)

Fig 17: Sibling Duo (Dedicated Cards Revealed)

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11. Notification Function & 12. Alarm Engine Mechanics

🔔 Notification & 🎵 Alarm Details:

- **Web Push (RFC 8291/8292):** Encrypted background alerts for join requests, absent notices, and make-up bookings.
- **Dual-Mode Audio Engine:** Pre-loads `a1arm.mp3` + Web Audio API 3-tone harmonic chime synthesizer fallback.
- **AudioContext Auto-Unlock:** Primed on first screen touch to bypass mobile browser auto-play restrictions.

13. Fee Structure & 10 Studio Instruments Catalog

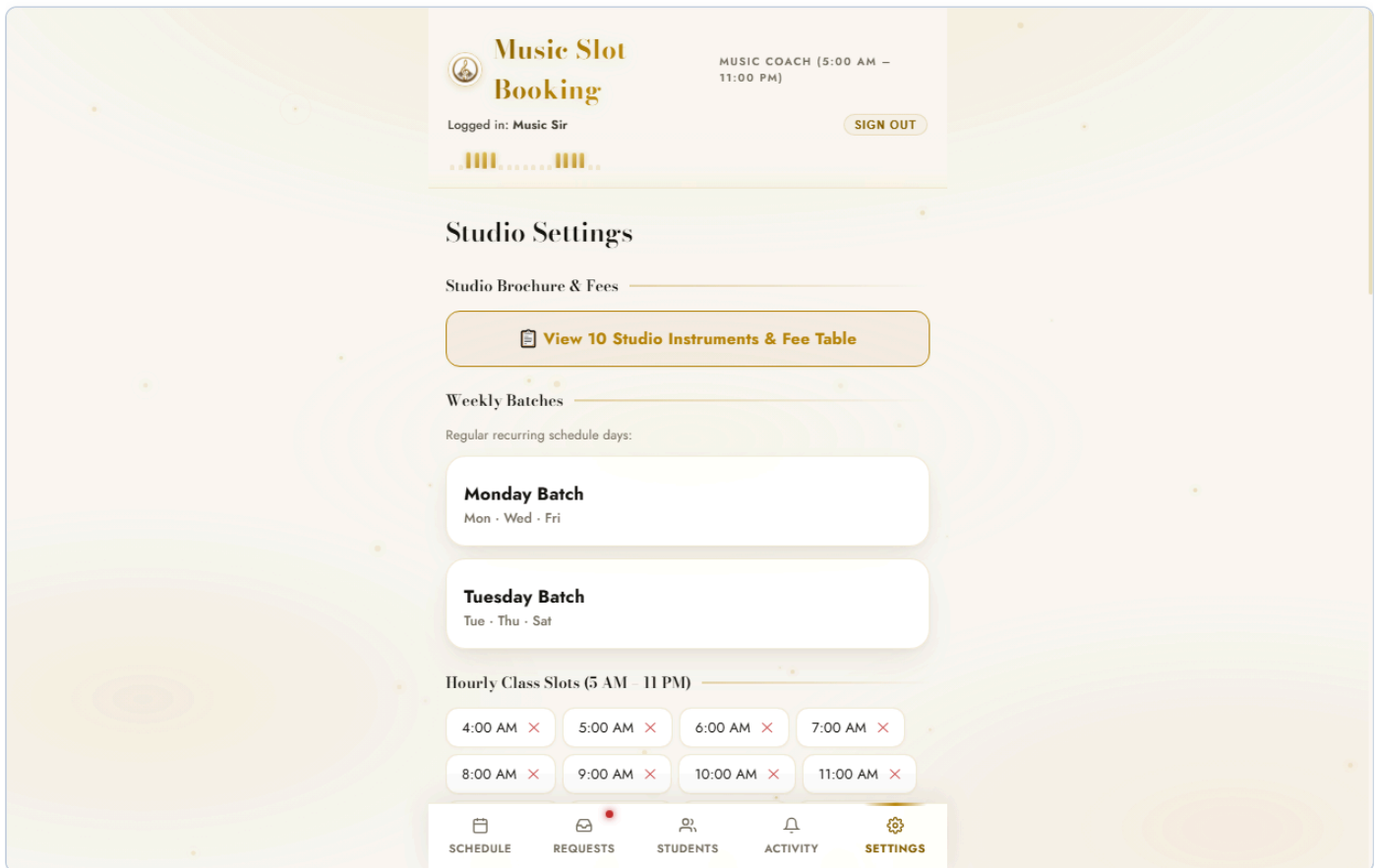


Fig 18: In-App Fee Structure & 10 Instruments Brochure

14. Database Persistence & 15. Cloud Deployment

Native SQLite in WAL mode ensures permanent, cross-device real-time sync between PC and mobile phone via Render / Railway.